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Vishwothama Nagar, Bantakal – 574115, Udupi District, Karnataka

Department of Artificial Intelligence and Data Science
Academic Year 2026-27

Phase 1: Software Requirements Specification Format

1. Title Page

Project Title:	**AI-Powered Smart Computer Laboratory Inventory Management System**
Team Members & USN:	**Thrisha K. Puthran** (USN: 4MW23AI058) *Department of Artificial Intelligence & Data Science*
Faculty Guide & Details:	**Guide Name:** Faculty Guide, Department of AI & DS **Designation:** Assistant Professor, Dept. of AI & DS **Institute:** Shri Madhwa Vadiraja Institute of Technology & Management
Target Facility:	AI & DS Academic Labs: Core Computing Lab (F106), Networks & Cloud Lab (F107), Edge Computing Lab, Server Facility
Academic Scope:	B.E. 5th/6th Semester Mini Project Phase-1

2. Introduction

Problem Statement – What problem are you solving?

In institutional computer laboratories such as SMVITM's AI & DS department, managing hundreds of capital workstations, digital ceiling projectors, high-speed network switches, and peripheral stock is currently performed through physical register books and uncoordinated spreadsheets. This manual procedure causes four major vulnerabilities: (1) Unauthorized or untracked equipment movement between labs without institutional oversight; (2) Delayed hardware defect reporting leading to prolonged student system downtime; (3) Time-consuming annual stock audits requiring manual serial number checks; and (4) Absence of an auditable two-tier approval chain connecting Lab Instructors, Department HOD, and the Principal.

Objective – What do you want to achieve?

- **Objective 1:** To build an autonomous web portal that tracks serialized workstations and bulk stock items across AI & DS laboratories in real time.
- **Objective 2:** To enforce a secure **Two-Tier Inter-Lab Transfer Approval Workflow** where equipment relocations require HOD endorsement and Principal executive sanction.
- **Objective 3:** To integrate native **QR Barcode Scanning** for instant bench inspection, continuous stock verification, and hardware specification retrieval.
- **Objective 4:** To embed a **Machine Learning Fault Diagnosis Engine** that categorizes defect descriptions and alerts technicians, alongside an institutional webmail dispatch system.

Scope – What will your system cover?

The system covers all capital equipment (177 high-performance workstations, Acer Veriton systems, Dell servers, projectors) and 900+ bulk inventory consumables (mice, keyboards, CAT6 patch cords, HDMI cables) in the AI & DS department. It provides role-based operations for Lab Instructor Chitrakala, HOD Tejaswini H, Principal Dr. Nagaraj Bhat, System Admin Thrisha, and hardware technicians. It operates continuously with automated stock movement logging and live notifications.

3. Proposed System

How the proposed system works:

The proposed system is an intelligent web application built using FastAPI (Python) and React 18. Each workstation is assigned a permanent tag (e.g. SMVITM-AIDS-F106-PC-01) and an encoded vector QR label. When an instructor requests equipment transfer to another lab, the system flags the item as 'Awaiting Approval', dispatches an email and in-app alert to HOD Tejaswini H. Upon HOD endorsement, an executive notification is routed to Principal Dr. Nagaraj Bhat. When the Principal sanctions the transfer, an instant celebratory neon popup alert and official email appear on the Lab Instructor's screen within seconds, clearing the hardware for dispatch. Faults reported by instructors are automatically categorized by an AI model and assigned to technicians.

Who will use it:

- **Lab Instructors (e.g. Chitrakala):** Bench management, QR scanning, fault logging, transfer initiation, and dispatch/receipt.
- **Head of Department (Tejaswini H):** Departmental inventory oversight, transfer endorsements, and lab utilization audits.
- **Principal (Dr. Nagaraj Bhat):** Executive institutional sanctioning, capital equipment audit reviews, and compliance oversight.
- **Hardware Technicians:** Diagnostic repairs, spare part logging, repair cost tracking, and maintenance resolution.
- **System Administrator (Thrisha):** User role management, database backups, audit trail inspection, and lab configurations.

Main benefits:

- **100% Traceability & Anti-Tamper Security:** Cryptographically tracked stock ledger eliminates untracked hardware relocations.
- **Instant Real-Time Communication:** Notification engine triggers live toasts and institutional emails within 3.5 seconds.
- **Accreditation-Ready Audits:** Generates VTU/NBA compliant stock verification CSV reports with one click.
- **Reduced Laboratory Downtime:** AI fault classifier accelerates maintenance ticketing and technician dispatch.

4. Functional Requirements

The functional requirements specify the exact capabilities and behaviors of the system:

FR1	The system shall allow authenticated users to log in securely using institutional email and role-based credentials.
FR2	The system shall allow lab instructors to register new workstations, generate standardized tags, and render vector QR codes.
FR3	The system shall allow users to scan physical QR codes via device webcams and display live workstation specifications and history.
FR4	The system shall allow lab instructors to report equipment faults and use an AI classifier to predict defect category and severity.
FR5	The system shall allow lab instructors to initiate inter-lab equipment transfer requests with mandatory justifications.
FR6	The system shall enforce a two-tier approval workflow requiring Department HOD endorsement and Principal executive sanction.

FR7	The system shall strictly prevent conflict of interest through anti-self-approval enforcement at both API and UI levels.
FR8	The system shall display an instant celebratory neon popup notification and dispatch an institutional email to the instructor upon Principal sanction.
FR9	The system shall allow technicians to update repair progress, record replaced spare parts, and log maintenance costs.
FR10	The system shall generate annual stock verification audit reports and export complete inventory CSV records for VTU/NBA compliance.

5. Non-Functional Requirements

- **Usability – Easy to Use:** The interface features an intuitive cyber-neon theme with a responsive Instructor Action Dock, clear typography, and a 60-second animated AI robot logout screen that guides users without cognitive overload.
- **Security – Authorized Access:** Enforces OAuth2 Password Bearer flow, bcrypt password hashing, JWT stateless tokens, strict role-based access control (RBAC), and backend anti-self-approval guards to prevent unauthorized privilege escalation.
- **Performance – Quick Response:** API response times remain under 200ms; client background notification polling operates at a 3.5-second interval; Vite bundle compiles in under 6 seconds with gzip compression.
- **Reliability – Data Integrity:** SQLite/PostgreSQL database enforces foreign-key referential integrity, atomic ACID transactions for stock balance deduction, and immutable audit logs for every state transition.

6. Modules

The system is divided into five cohesive modules with defined actors and features:

1. Authentication & RBAC Module **Actors:** All Users (Instructor, HOD, Principal, Tech, Admin)	Handles secure user login, password hashing, JWT issuance, session expiry, role-based dashboard rendering, and user profile management.
2. Smart Asset & QR Inventory Module **Actors:** Lab Instructor, Admin	Manages 177 capital assets and 915 bulk items; generates unique asset tags; encodes printable QR labels; tracks live specifications and allocation across 6 labs.
3. AI Fault Detection & Ticketing Module **Actors:** Lab Instructor, Technician	Provides natural language defect logging; executes Scikit-learn TF-IDF machine learning inference to classify defect category; tracks repair lifecycle and spare part costs.
4. Two-Tier Transfer Approval Module **Actors:** Instructor, HOD, Principal	Facilitates inter-lab equipment movements; enforces sequential HOD endorsement and Principal executive sanction; prevents self-approvals; triggers celebratory popups and emails.
5. Stock Audit & Webmail Reporting Module **Actors:** Auditor, HOD, Principal, Admin	Enables rapid bench QR scanning; compares physical items with digital ledger; flags discrepancies; hosts an Institutional Webmail Inbox; exports official CSV audit reports.

7. System Diagram (Overall Block / Architecture Diagram)

Simple Block Architecture Diagram as per College Specification:

User → Frontend (React 18) → Backend (FastAPI) → ML Model / Database → Result / Action

8. Use Case Diagram

Main Actors and Functions per Module:

9. Sequence Diagrams for Each Major Module / Functionality

Chronological message flows across modules:

A. Two-Tier Inter-Lab Transfer Approval & Real-Time Notification Flow

B. AI-Assisted Fault Logging & Maintenance Resolution Flow

10. Technology Used

Component	Technology Selected	Purpose / Role in Architecture
Frontend	React 18, Vite, Tailwind CSS, Lucide Icons	Responsive UI, Instructor Dock, audio chimes, modal workflows
Backend	Python 3.13, FastAPI, Pydantic v2	High-performance REST API, asynchronous validation, RBAC security
Database	SQLite (inventory_smvitm.db) / PostgreSQL	Relational persistence of 177 assets, ledger logs, foreign keys
Machine Learning	Scikit-learn, TF-IDF Vectorizer, Joblib	Natural language fault classification and severity prediction
QR Engine	qrcode (Python), html5-qrcode (JS)	Vector barcode generation and client webcam camera scanning
Communication	SMTP Gateway + In-App Polling (3.5s)	Institutional email dispatching to principal@sode-edu.in & alerts
Security	Passlib (bcrypt), PyJWT (HS256)	Password hashing, stateless bearer authentication, anti-self-approval

11. Expected Outcome

The completed AI-Powered Computer Laboratory Inventory Management System will deliver an end-to-end institutional platform for the Department of Artificial Intelligence and Data Science at SMVITM Bantakal. It successfully eliminates manual paper registers and untracked asset displacement by establishing complete digital visibility over all 177 capital workstations and 900+ peripheral accessories across 6 departmental computer facilities.

By enforcing a secure, two-tier transfer approval workflow (HOD endorsement → Principal executive sanction) paired with anti-self-approval validation and instant real-time notification dispatch, the system guarantees 100% accountability and compliance with VTU autonomous academic guidelines and NBA accreditation requirements.

Comments and observation by guide:

On regularity: _____

On work progress:

_____

Mini Project coordinator Remarks and Signature:

_____

Date: _____

HOD Signature:

_____

Date: _____